

SciForge: A Multimodal Research-Native AI Workbench for Scientific Discovery

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Abstract

Scientific work increasingly spans heterogeneous artifacts—papers, code, datasets, scientific file formats, model outputs, figures, manuscripts, and team decisions—yet general-purpose AI assistants rarely preserve these objects as a coherent, auditable research state. We present SciForge, a multimodal research-native AI workbench that reserves the graphical interface for human judgment while search, parsing, model routing, workflow execution, plotting, writing, and presentation generation run as modular agent-accessible services. SciForge is built around five pillars: (i) *goal-scoped scientific decision governance* for **goal-oriented** research, with review gates and shared review surfaces; (ii) *translate-then-reason* for **multimodal** input, routing scientific objects through domain translators before the agent reasons; (iii) *evidence governance* for **auditable** traceability, linking claims to provenance chains and audit findings; (iv) *collaborative team science* for **collaborative** research, enabling multi-role decision governance, with shared team workspaces planned for future releases; and (v) *real-world application scenarios* for **practical** impact, integrating end-to-end scientific workflows including BGC discovery, protein structure prediction, automated figure generation, and multi-format document parsing. The system combines a thin interaction layer, contextual research capability patterns, an Agent Runtime and Workflow Engine, an Evidence-DAG audit sidecar, a Scientific Model Router, and local scientific memory backed by reusable worker services. SciForge currently runs as a desktop application, with mobile supervision support; future releases will deepen team collaboration through shared review surfaces and decision governance. The system is open-source and available at <https://github.com/AGI4Sci/SciForge>.

Keywords: SciForge, AI workbench, scientific discovery, agentic workflow, evidence-aware AI, multimodal science, research automation

1 Introduction

1.1 Background and Motivation

Scientific discovery is becoming increasingly computational, multimodal, and collaborative. A single research project may move among scholarly papers, experimental protocols, command-line tools, notebooks, gene and variant files, protein sequences, three-dimensional structures, molecular records, single-cell matrices, generated figures, manuscript drafts, slide decks, and group decisions. These objects are not independent files. A manuscript claim may depend on a raw dataset, a preprocessing script, a model run, a visualization parameter, a literature passage, and a later human decision about whether the evidence is strong enough to use. As a result, the central bottleneck is no longer only access to models or tools, but the lack of a persistent environment that can organize scientific objects, agent actions, human decisions, and evidence traces as one coherent research state.

Recent AI systems have demonstrated impressive capabilities for literature search, hypothesis generation, coding, experimentation, and scientific writing [9–12, 19]. At the same time, desktop and terminal-capable scientific workbenches show the importance of keeping execution close to local files, remote servers, and HPC environments [4, 6, 13, 16, 20]. However, most existing systems still emphasize either task-specific intelligence, conversational assistance, or local execution surfaces. They rarely treat the full research process as a **goal-oriented**, **multimodal**, **auditable**, and **collaborative** workflow in which scientific data, agent reasoning, generated artifacts, and manuscript claims remain connected over time.

This gap matters because scientific AI agents must satisfy four demands that are absent from general-purpose assistants. First, **goal-oriented** research design is indispensable: scientific work is goal-scoped, not session-scoped—researchers need persistent objectives, review gates, and shared decision records that span turns, tools, and collaborators. Second, **multimodal** inputs are the norm: scientific files arrive in native representations poorly suited to direct prompting, including FASTA sequences, PDB or mmCIF structures, SMILES strings, variant files, and cell-omics tables. Third, **auditable** evidence is non-negotiable: researchers need to know which data, script, model, parameter, paper, or expert translation supported a conclusion; whether a figure matches the claim it is used to support; and when a proposed action requires human or principal-investigator approval. Fourth, **collaborative** governance is essential: scientific work involves team decisions, goal-scoped approvals, cross-session handoffs, and shared review of claims and evidence. A stateless chat transcript addresses none of these demands.

We present *SciForge*, a multimodal research-native AI workbench for scientific discovery. SciForge is designed as a local-first operating environment for research rather than a wrapper around a single model. Its architecture organizes interaction surfaces, research capability patterns, a core engine with evidence governance, a scientific model router, and local-first infrastructure into a coherent layered workbench; Section 3 describes each layer in detail.

1.2 Contributions

SciForge’s core scientific contribution is organized around five pillars, each addressing a distinct gap in current research AI systems:

- **Native scientific object processing with translate-then-reason:** end-to-end file ingress exposes four classes—protein sequences (.faa; .fasta/.fa with per-record conservative content classification), protein structures (.pdb/.cif/.mmcif), molecules (.smi/.smiles), and single-cell transcriptomics (via C2S worker contract)—routed through domain-expert translators (Esm2Text [1], Prot2Text [1], BioT5 [17], and C2S [18]) that return structured expert observations before the main agent reasons. The translator output is an evidence candidate, not a verified scientific fact; the human remains responsible for judgment.
- **Artifact- and run-grounded evidence governance:** every agent action—file read, model call, tool invocation, script execution—can attach structured provenance (software version, parameters, environment, random seed) to a thread-level Evidence Snapshot. Users designate when to commit a snapshot, marking which evidence supports which claim. A read-only audit sidecar inspects these snapshots and surfaces potential gaps without blocking the agent’s work.
- **Goal-scoped scientific decision governance:** a Project DAG organizes research goals, evidence snapshots, and review decisions into a shared, auditable structure. Researchers can inspect evidence chains, record review decisions, set goal-level execution policies (autonomous, checkpointed, or supervised), and promote artifacts through candidate and certified release gates.
- **Collaborative team science:** SciForge supports multi-role decision governance with reviewer/approver roles and IM-based coordination channels (Zulip, Discord, WeChat, Feishu) for remote task submission and status supervision. The Project DAG enables collaborative review workflows where researchers inspect Evidence Snapshots, record ReviewItems with role-attributed decisions, and advance artifacts through candidate/certified release gates. Shared team workspaces with cross-session research memory synchronization are planned for a future release.
- **Real-world scientific application scenarios:** SciForge integrates end-to-end scientific workflows for several concrete research tasks, including biosynthetic gene cluster (BGC) discovery from genomic data, protein structure prediction and analysis via AlphaFold3 integration, automated scientific figure generation with journal-style rendering, multi-format document parsing for scientific literature ingestion, and AI-assisted presentation slide deck generation from research artifacts. These applications demonstrate the system’s ability to bridge the gap between general-purpose AI assistance and domain-specific scientific practice.

SciForge treats each scientific interaction—literature triage, hypothesis refinement, manuscript drafting, artifact inspection, and presentation—as a first-class research task with explicit evidence trails, audit findings, and governance checkpoints. The SciForge codebase, including the Evidence-DAG audit worker, decision-record contracts, agentic skills, and scientific model router, is open source at <https://github.com/AGI4Sci/SciForge>.

2 Related Work

2.1 GUI-Based Scientific Workbenches

GUI-based scientific workbenches, including desktop and web applications, make AI assistance accessible through human-facing workspaces. Claude Science is the closest broad commercial reference: it is described as an AI workbench for scientists with desktop, local, SSH, and HPC-facing execution, scientific skills, specialist agents, generated artifacts, and reviewer agents for citations and calculations [4]. ClaudePrism focuses on the manuscript side of the workflow, combining a native desktop environment with local LaTeX compilation, Python environments, Claude Code sessions, and scientific writing skills [6]. Web tools such as Elicit and Consensus provide polished interfaces for literature search, synthesis, and research-agent style review [7, 8]. These systems demonstrate the importance of usable research interfaces, but they are usually centered on one model family, one workflow domain, or one user session. SciForge instead treats the GUI as a thin layer for human judgment over a local, modular, multi-agent research substrate.

2.2 Terminal Workbenches

Terminal-oriented workbenches prioritize execution close to local files, remote servers, and HPC environments. OmicOS provides a Rust `omicos-core` binary that can run as a local daemon serving a browser UI or as `omicos cli`, with analysis code running in a shared IPython kernel and raw data remaining local [13–15]. OmicsClaw follows a local-first multi-omics design with CLI, TUI, desktop, web backend, and SSH remote execution surfaces over a shared agent loop [16]. Operon targets bioinformatics workflows across desktop and remote cluster environments, using persistent terminal sessions so jobs execute where data and schedulers live [20]. These systems are strong at local and remote execution, but they generally expose tools and workflows rather than a unified layer for multimodal scientific routing, research decision surfaces, team supervision, and manuscript-to-artifact continuity.

2.3 API and Research Agents

API-based and research-prototype agents explore more autonomous scientific reasoning. FutureHouse exposes agents for literature search, deep review, precedent search, and chemistry planning through a web/API platform [10]. Google’s AI Co-Scientist studies multi-agent hypothesis generation, debate, and ranking for biomedical research [11]. The AI Scientist and Agent Laboratory automate parts of the machine-learning research loop, including ideation, coding, experimentation, paper drafting, and review [12, 19]. Literature systems such as PaperQA2, OpenScholar, and ScholarQA focus on retrieval-augmented scientific question answering and literature synthesis [2, 5, 9]. These systems demonstrate the power of specialized agents, but they are less focused on being a persistent local workbench where data, code, figures, decisions, and generated artifacts remain in one workspace.

2.4 Comparison and Positioning

While each of the three system categories addresses a distinct facet of the AI-assisted research life cycle, SciForge is designed to unify them within a single workbench. The comparison spans five key dimensions: execution, model governance, evidence quality, research continuity, and goal-oriented research design.

Execution architecture. GUI workbenches such as Claude Science and Claude-Prism deliver polished interactive experiences but are typically constrained to one model provider and one local session. Terminal workbenches—OmicOS with its hybrid daemon plus browser model, OmicsClaw with multi-surface CLIs, and Operon with persistent terminal sessions—excel at running code where data and schedulers reside, yet they expose tools and pipelines rather than a unified research workbench. SciForge bridges these worlds by providing a GUI for human judgment while keeping a local-first workspace where execution can span desktop, SSH, and HPC environments.

Model governance. The reviewed public materials commonly emphasize a single provider family or manual switching between separate tools. SciForge’s Scientific Model Router combines multi-provider routing with a deliberately narrow translate-then-reason file-ingress path for protein sequences, protein structures, and molecules. C2S remains a worker API contract rather than an end-to-end file route, and unsupported protected formats fail closed. Literature is handled through search, PDF anchoring, and source attribution, not as a scientific modality translator. We did not find this same integrated combination documented as a first-class feature in the reviewed categories.

Evidence quality and governance. Research agents such as FutureHouse, AI Co-Scientist, and the AI Scientist produce outputs from variable-quality evidence; in the publicly documented materials we reviewed, we did not observe the same integrated combination of claim–evidence audit trail, PROV-JSON provenance, and quality gates in a single research workbench. SciForge threads claims through an Evidence DAG backed by PROV-JSON provenance, audit runs, risk digests, and explicit quality gates. This governance layer is designed to make agent outputs inspectable and auditable.

Research continuity. Many reviewed systems foreground per-session interaction. SciForge maintains goal-centered research memory as scoped free-text records (scope, tags, confidence, timestamps, context) across sessions; typed goals and decision events belong to the Project DAG, enabling long-running multi-session scientific workflows rather than only single-turn conversations. The combination of research memory, workflow engine, and evidence governance creates a persistent research workspace whose integrated design—tying thread-level Evidence Snapshots to goal-scoped Project Snapshots with audit and release semantics—we did not find documented as a first-class, integrated feature in the public materials of the reviewed GUI, terminal, and API-agent systems.

Goal-oriented research design. Beyond per-task execution, SciForge anchors every thread and project to an explicit goal scope with defined review gates, decision records, and release semantics. This **goal-oriented** design distinguishes SciForge from systems that treat each conversation as an independent session: scientific objectives in SciForge are persistent, auditable, and governable across multiple turns, tools, and contributors. The combination of goal-centered memory, the Project DAG, and shared

review surfaces gives research teams a concrete mechanism to track progress toward scientific milestones rather than accumulating disconnected interactions.

Table 1 summarizes these differences at the category level, while the detailed capability differentiation in Appendix A1 provides a feature-level comparison against common approaches.

Table 1 High-level comparison of scientific AI system categories. A checkmark indicates first-class or documented support as a primary design goal in the publicly reviewed materials. Absence of a checkmark does not prove absence of a feature; it indicates we did not find documented first-class support in the reviewed materials.

Dimension	GUI WB	Terminal WB	API agent	SciForge
Human-facing UI	✓		✓	✓
Local/HPC execution		✓		✓
Scientific tools	✓	✓	✓	✓
Local-first workspace		✓		✓
Multi-provider routing				✓
Scientific multimodal routing				✓
Human decision surfaces	✓			✓
Team workspace sync				✓
Modular worker services		✓		✓

SciForge combines these categories rather than competing with only one of them. It keeps the usability of GUI workbenches, the locality and composability of terminal systems, and the task power of scientific agents, while adding fail-closed scientific-object handling, content-addressed artifacts with structured selection and provenance, specialist domain translation, run-grounded evidence governance with immutable snapshots and an audit sidecar, and goal-scoped decision governance with explicit release semantics.

3 System Architecture

3.1 Overall Design Philosophy

SciForge is organized as a layered research operating environment. **Layer 1** provides thin judgment surfaces for desktop, group, and mobile interaction. **Layer 2** encapsulates six research capability patterns—literature synthesis, data analysis, hypothesis design, reproduction, writing, and review—that share an evidence-aware control chain. **Layer 3** houses the core engine: agent runtime, workflow automation, scoped memory, and evidence governance. **Layer 4** is the Scientific Model Router that detects modality and dispatches to domain-expert translators. **Layer 5** supplies infrastructure: local memory, modular workers, scientific connectors, and reproducible run substrate.

SciForge follows three implementation principles that distinguish it from feature-heavy desktop software:

1. **Thin GUI.** The interface appears only when human judgment is needed—reviewing, annotating, approving, comparing; all other capabilities run in the background agent runtime.
2. **Modular services.** Search, parsing, plotting, model routing, workflow execution, and presentation generation run as MCP-compatible tools or worker processes, keeping the application close to a thin interaction shell [3].
3. **Local-first evidence model.** Data, traces, artifacts, and evidence graphs remain anchored to the research workspace; remote collaboration and mobile access are control channels, not replacements for local state.

Figure 1 summarizes the layered architecture.

3.2 User Interaction and Collaboration (Layer 1)

Layer 1 exposes three interaction surfaces driven by the same underlying runtime. The **desktop application** provides a thin GUI reserved for human judgment—reviewing evidence, annotating figures, comparing candidate artifacts, and approving release decisions—while execution runs in the background. **Group chat integration** (Zulip, Discord, WeChat, Feishu) enables remote task submission, status polling, and lightweight approvals without switching context. **Mobile supervision** offers read-only monitoring of long-running experiments and review tasks, with interaction gated through the same governance model.

3.3 Research Capability Patterns (Layer 2)

Research capability patterns expose six recurrent scientific operations with deliberately distinct boundaries. **Literature Synthesis** maps papers and source records to an evidence summary. **Data Analysis** maps datasets and analysis parameters to quantitative results and figures. **Hypothesis Design** maps existing evidence and a research goal to testable hypotheses; acceptance or rejection is not part of this operation and remains a Project DAG decision. **Reproduction** maps a prior paper, code, data, and protocol to an independently rerun result. **Research Writing** maps evidence and research artifacts to manuscript text. **Visual Review** maps figures, slides, or visual artifacts to annotations and candidate revisions. These are contextual patterns within one workbench, not independent applications with separate execution semantics. Regardless of entry point, each task follows the same evidence-aware control chain: (1) task and object ingress; (2) fail-closed translation when a native scientific object requires a domain expert; (3) delegated agent and worker execution through Layers 3 and 4; (4) provenance-attached evidence candidates with artifact and run lineage; (5) a thread-scoped Evidence Snapshot and advisory audit; and (6) goal-scoped review and candidate/certified release through the Project DAG and human PI.

Two complementary DAG substrates implement this shared pattern. The **Evidence DAG** is thread-scoped: each agent thread grows its own claim–source–reasoning graph that records the evidentiary structure of that conversation’s reasoning. The **Project DAG** is goal-scoped: it aggregates auditable evidence across threads into a project-level graph, linking claims, audit findings, and decision records to the objectives they support. Audit runs remain advisory and fail open during exploration, while

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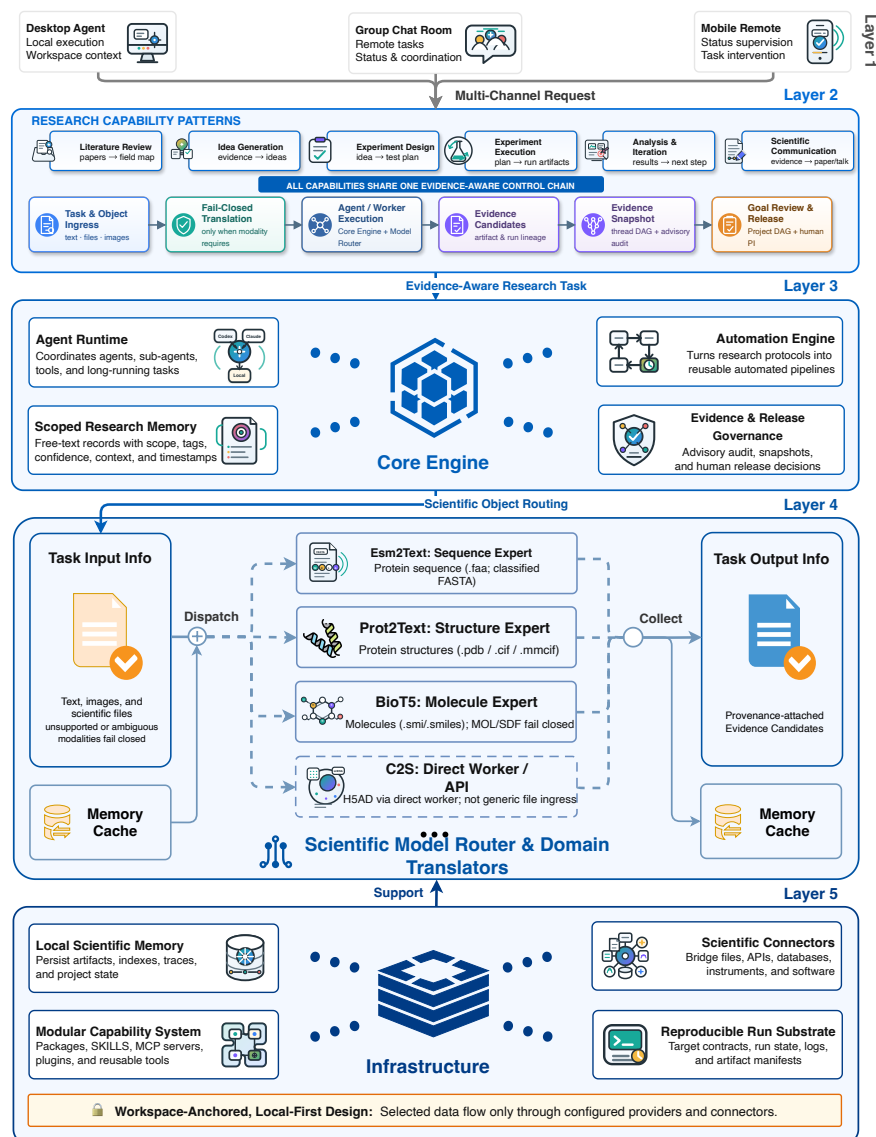


Fig. 1 SciForge system framework. Six research capability patterns share a common evidence-aware control chain—ingress, translation, execution, evidence capture, audit, and release—built on the layered architecture of interaction surfaces, core engine, scientific model router, and local-first infrastructure.

lightweight decision records capture explicit human judgments such as artifact review, acceptance or rejection, and release actions. Background work—search, PDF parsing, data exploration, plotting, protocol drafting, reproduction pipelines, manuscript and presentation generation, and workflow execution—is delegated to the Agent Runtime and workers, keeping the primary workbench surface available for human judgment.

3.4 Core Engine (Layer 3)

The Core Engine plans and executes research work through four coordinated subsystems. The **Agent Runtime** uniformly supports multiple agent backends—Codex, Claude Code, and custom runtimes—allowing research groups to select the execution environment that best matches their scenario: Codex for local-first interactive coding with full workspace access, Claude Code for extended autonomous task chains, and custom runtimes for specialized scientific computing or institutional deployment requirements. Across all backends, the Runtime coordinates the main agent, delegated sub-agents, tool execution, workspace operations, MCP worker invocation, and long-running task recovery through a consistent API surface. The **Automation Engine** formalizes reusable pipelines with code, agent, tool, and approval nodes; scheduled execution and loop-style fixed protocol automation are treated as run modes rather than separate architectural modules. **Scoped Research Memory** stores free-text records (scope, tags, confidence, timestamps, context) across sessions; typed goals and decision events belong to the Project DAG. **Evidence & Release Governance** captures provenance and claim–evidence structures, creates reviewable snapshots and audit findings, and exposes goal-scoped release decisions without synchronously blocking exploratory execution by default. Multimodal translation caches store scientific and visual observations by content hash, enabling repeated requests to reuse expert evidence without recomputing final conclusions.

Evidence DAG and Evidence Audit Runs. The Evidence-DAG sidecar builds a thread-scoped claim–evidence graph from completed agent turns. The runtime normalizes user messages, assistant/reasoning items, and successful tool results into a trace feed, then posts completed-turn deltas that grow the thread graph by default; explicit rebuilds are reserved for whole-thread reconstruction. The sidecar extracts source, reasoning, and claim nodes; verifies newly added support and contradiction edges when a Model Router judge is available; persists the graph as PROV-JSON; and runs a lightweight deterministic audit by default. Verification and automatic audit are advisory and fail open, so they do not block the agent’s exploratory work. The Workbench Evidence DAG panel can also manually build and audit the current thread: the manual action backfills the active thread through the same trace feed, runs an audit over the current DAG, refreshes the embedded view, and reports a risk digest. Audit runs do not mutate the graph. They produce structured findings—for example ungrounded claims, unresolved contradictions, weak support, single-source dependencies, hidden shared-source support, load-bearing evidence, or low-credibility sources—which can later inform human decision records, workflow approval gates, or pre-publication commit checks.

3.5 Scientific Model Router (Layer 4)

The **Scientific Model Router** is a core architectural innovation of SciForge, acting as a universal model invocation plane that abstracts away provider heterogeneity while retaining the full expressiveness of each backend. Rather than forcing a lowest-common-denominator interface, the Router supports three request–response protocols—OpenAI Chat Completions, OpenAI Responses, and Anthropic Messages—selected per provider

through a normalized endpoint format, allowing each model to be invoked through its native, most capable API surface.

Intelligent Auto-Routing. The Router includes a lightweight classifier that inspects the user request and recent context to automatically select the appropriate reasoning effort (off, high, or max), with a deterministic heuristic fallback so routing decisions do not rely on a single point of failure.

Provider-Aware Optimization. For DeepSeek models, the Router performs proactive reachability probing, applies native thinking controls, and computes real-time cache-aware cost estimation in both USD and CNY. For all providers, it implements automatic retry with backoff, error classification, and stream-idle timeout detection.

Output Robustness and Repair. The Router repairs malformed tool-call arguments—stripping Markdown fences and recovering partial JSON—and extracts embedded images from tool results as standard model attachments, enabling multimodal reasoning chains across scientific visualizations and model analysis.

Scientific Modality Routing. For scientific files, the Router detects modality-specific extensions and applies one of two handling paths. *Translate-then-reason* applies to four currently supported modalities: protein (.faa; .fasta/.fa with per-record conservative content classification) translated by Esm2Text, protein structures (.pdb, .cif, .mmCIF) by Prot2Text, small molecules (.smi, .smiles) by BioT5, and single-cell transcriptomics via the C2S worker contract. Each translator output carries provenance metadata, allowing the main agent to reason over structured expert observations. *Code-to-Science (C2S)* is a planned worker API contract that will allow code execution outputs to be routed through the same translate-then-reason pipeline, enabling computational results to participate in scientific reasoning chains. VCF, BED, GFF, and MGF are unsupported and fail closed; raw content is not routed to the general reasoner.

3.6 Collaboration and Team Coordination (Supported)

SciForge’s current collaboration model centers on the desktop application with local-first workspace sharing through version-controlled project directories. Supported extensions provide real-time shared review surfaces for evidence inspection, multi-PI decision governance through the Project DAG, and IM-channel integration (Zulip, Discord, WeChat, Feishu) for remote task submission and status coordination. The **goal-oriented** research design means that team members operate within shared goal scopes: each scientific objective carries explicit review gates, evidence snapshots, and decision records that are visible to all collaborators, enabling transparent, auditable team science rather than isolated individual sessions.

3.7 Infrastructure (Layer 5)

The Infrastructure layer provides the durable substrate that anchors research activity to the local workspace. **Local Scientific Memory** stores papers, datasets, parsed documents, scientific file manifests, embeddings, indexes, tool outputs, run metadata, and artifact caches. The **Modular Capability System** packages computational functionality as Skills, MCP servers, Plugins, HTTP services, or worker processes—each independently versioned and upgradeable. **Scientific Connectors** normalize

external resources such as papers, datasets, databases, APIs, instruments, notebooks, analysis scripts, reference managers, LIMS/ELN systems, and institutional storage into a unified access layer. The **Reproducible Run Substrate** records execution-target contracts, run and job state, logs, and artifact manifests so that downstream evidence can refer to concrete execution records. SciForge remains workspace-anchored and local-first; selected data may flow through explicitly configured model providers, connectors, or remote execution services.

4 Use Cases

This section uses three evidence-status labels consistently: *[Implemented]* for a product path exercised in SciForge, *[Validated Prototype]* for a bounded single-case or externally archived pilot with stated limitations, and *[Illustrative Future]* for an unexecuted target workflow. These cases are not separate product panels; they show what is implemented, what has only pilot evidence, and what remains a design target on the same modular substrate.

Fig. 2 Classification of AI-for-Life-Science scenarios represented by SciForge use cases. The labels distinguish an exercised product path (Implemented), bounded pilot evidence (Validated Prototype), and unexecuted design targets (Illustrative Future); they do not imply equivalent validation across cases.

Scientific scenario	Representative workflow	Evidence status
Literature-driven discovery	Agentic research sprint: meiosis initiation switch (scenario-01)	[Implemented]
Automated AI modeling	AI4AI-driven cell perturbation prediction with iterative algorithm-atom recombination	[Validated Prototype]
Evidence-governed writing	Reviewer/rebuttal mode with a claim-evidence response matrix	[Validated Prototype]
Paper reproduction	MCFST spatial-transcriptomics guided reproduction	[Validated Prototype]
Multi-source integration	Cross-scale cell atlas spanning ECCITE-seq and external resources	[Illustrative Future]
Molecular discovery	AI-guided protein design for stability, specificity, or <i>de novo</i> binding	[Illustrative Future]
Molecular design and optimization	MolClaw agent-driven EGFR inhibitor design with 6-round docking optimization	[Validated Prototype]
Lab automation	Genome-to-BGC discovery and prioritization with multi-Agent scientific analysis	[Validated Prototype]

4.1 Agentic Research Sprint: From Question to Manuscript Package

Setting. *[Implemented]* A research group—led by a human PI—poses an open-ended biological question, “*Which molecular systems drive mammalian germ cells into meiosis?*”, and defines the research requirements and stage-gates. The PI then delegates execution to Codex, an AI coding agent that acts on the PI’s behalf: Codex repeatedly invokes SciForge’s agent-accessible services—literature search, candidate gene identification, structured hypothesis formulation, and parallel sub-agents for multi-omics

evidence synthesis, AlphaFold3 structure-based computational assessment, and iterative methodology hardening—within a multi-day, PI-controlled agentic loop (Fig. 3). We emphasize that Codex serves as the PI’s AI delegate and does *not* replace SciForge; SciForge remains the research workbench whose modular services (search, routing, workflows, plotting, writing) the agent consumes. Selected outputs are reviewed through evidence summaries, reviewer-style self-audits, and human decisions; the PI tightens evidence boundaries and repairs tooling as needed. After 132 stages and 197 Git-tracked commits, the repository¹ records the control loop as a local workspace log: hypotheses, evidence ingestion, structural analysis, manuscript integration, and reviewer-style self-audits.

Discovery. The systematic agentic pipeline yielded a candidate-gene atlas of 23 genes across 12 molecular classes, with claims traceable to specific evidence sources (Fig. 4). Central to the discovery was the DCT-M3 causal-module triangulation framework, which decomposes the meiosis initiation switch into four regulatory layers: (1) permissive licensing—chromatin opening and epigenetic priming via pioneer transcription factors; (2) brake release—degradation of meiotic inhibitors through ubiquitin-proteasome pathways; (3) trigger execution—SPO11-mediated programmed double-strand breaks and synaptonemal complex assembly; and (4) chromatin/RNA competence—transcriptional activation, alternative splicing, and 3D genome reorganization. Multi-omics evidence (transcriptomics, epigenomics, proteomics) was cross-validated against AlphaFold3 structure-based computational assessments and systematic literature triangulation to ground each candidate in causal evidence. This scenario demonstrates that SciForge can sustain long-horizon, evidence-constrained research—not by auto-generating a paper, but by producing an inspectable, auditable research package whose reasoning and evidence trail remain open to human scrutiny.

Informal Post-Hoc Literature Mapping. A post-hoc review of the 23-gene atlas against the published meiosis literature found that 22 of the 23 genes have known meiosis associations; MAPK fell outside established meiosis paradigms. This retrospective mapping was conducted informally and was not guided by a pre-registered protocol. Formal expert adjudication of gene-level relevance, along with inter-annotator agreement metrics and a pre-registered evaluation protocol, remains future work.

¹<https://github.com/AGI4Sci/scenario-01-research-sprint> (archive currently unavailable; the URL returned 404 at time of writing)

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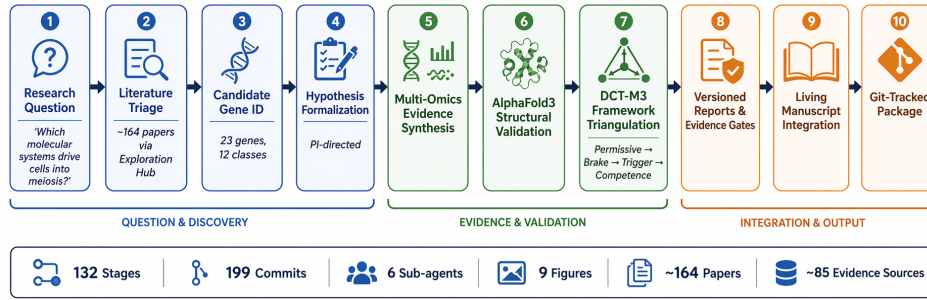
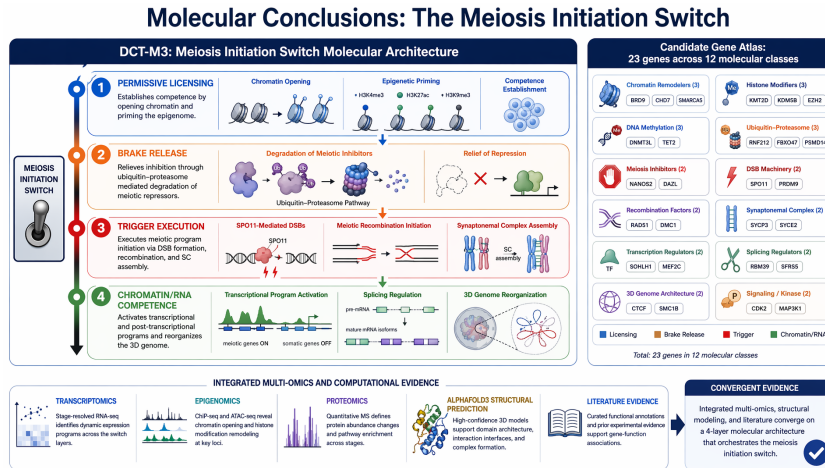


Fig. 3 Automated research pipeline executed during the SciForge scenario-01 research sprint. The 132-stage, 197-commit agentic loop progresses from a biological research question through literature mining (DCT-M3), candidate gene curation, AlphaFold3 structural prediction, evidence triangulation, manuscript synthesis, and submission packaging. Quantitative metrics: 132 research stages, 197 Git commits, 6 parallel child-agent runs, 9 generated figures, ~164 papers reviewed, and ~85 evidence nodes produced. Selected artifacts and thread evidence are linked to versioned records, evidence summaries, and audit findings where captured.



4.2 AI4AI: Automated Modeling for Scientific Discovery

[Validated Prototype] AI-for-AI (AI4AI) refers to a design paradigm in which agents read, distill, and recombine algorithmic building blocks from existing codebases into new machine learning models. Rather than treating model design as a one-shot prompt, AI4AI treats it as an iterative design loop: an agent reads external repositories, extracts reusable algorithm atoms (e.g., attention mechanisms, distribution losses, latent regularizers), maps them to the target project’s training interface, composes a new model, and submits training, inference, and evaluation jobs through a computing-platform job-submission system. Evaluation metrics then feed back into the next design cycle.

We exercised this paradigm on cell perturbation prediction—the task of predicting how a cell’s transcriptome changes in response to a genetic or chemical perturbation—using a NeMo/BioNeMo training stack supporting PBMC, Replogle, and Tahoe benchmark datasets. The agent distilled algorithm atoms from the `stack` single-cell foundation model repository, producing structured design cards covering Cell-by-Gene Tabular Attention, Rectangular Gene Masking, Distribution Heads, Sliced Wasserstein Regularization, and In-Context Generation. These atoms were mapped to the project’s NeMo model interface (MInterface, DInterface) and composed into new perturbation-prediction architectures under a frozen RJOB submission contract that forbids ad-hoc code changes during job submission.

Delta PCC—the improvement in Pearson correlation over a baseline JIT model—was tracked across design iterations. In the recorded pilot, Delta PCC improved from 0.30 (Iteration 1) to 0.48, 0.58, and finally 0.64 (Iteration 4), demonstrating that iterative atom-level design recombination can yield measurable performance gains (Fig. 5). The complete agent logs, design cards, model code, configurations, and evaluation records are archived at <https://github.com/BruthYU/CELL-AI4AI>.

Limitations: The Delta PCC curve represents a single pilot trajectory with four design iterations; statistical significance, cross-seed stability, and generalization to new perturbation tasks have not been established. The algorithm-atom distillation quality depends on the agent’s reading fidelity and the target repository’s code organization. The current pilot uses post-hoc metric reporting without pre-registered stopping criteria or blinded evaluation. Formal validation would require multi-seed replication, pre-registered success thresholds, and independent re-execution on held-out perturbation types.

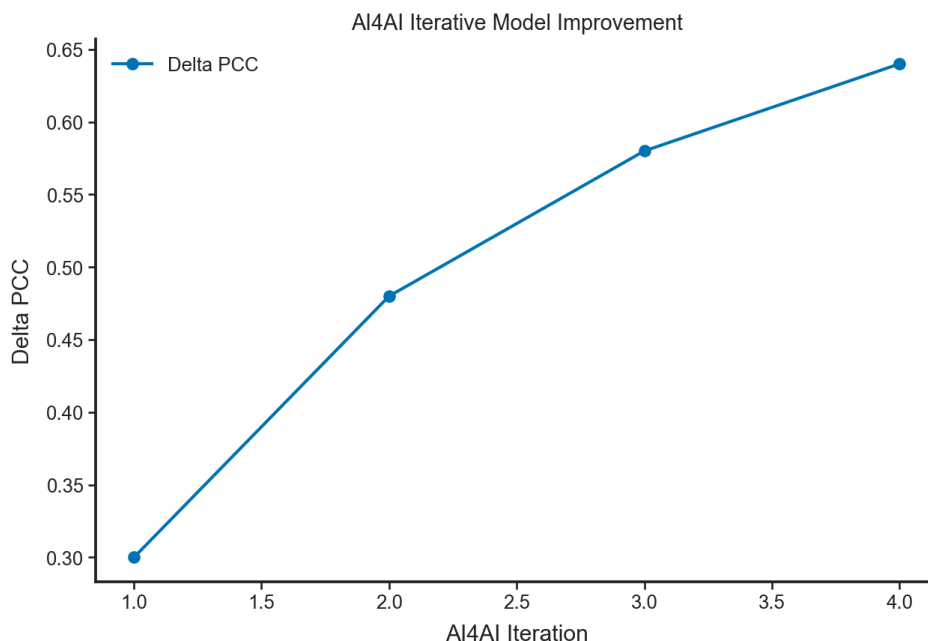


Fig. 5 Delta PCC improvement across AI4AI design iterations for cell perturbation prediction. The curve tracks iterative algorithm-atom recombination on PBMC, Replogle, and Tahoe datasets under a NeMo/BioNeMo training stack.

4.3 Reviewer / Rebuttal Mode: Evidence-Governed Peer Review and Response

Scenario. *[Validated Prototype]* To exercise SciForge’s reviewer/rebuttal capability, we conducted a structured six-stage sprint against the manuscript “*Benchmarking Virtual Cell Models for In-the-Wild Perturbation Response*” (arXiv:2604.27646v1, VCBench) [21]. This is a one-manuscript SciForge-assisted first pass; real reviewer comments, follow-up analyses, and manuscript revisions are not yet completed. The full sprint log, outputs, and SciForge GUI trace are archived in a dedicated repository.²

Stage 1—Intake. The manuscript PDF and LaTeX source were frozen and ingested. Figures, tables, datasets, and code references were catalogued as evidence objects.

Stage 2—Claim Extraction. Contribution claims, result claims, limitation claims, and practical-guidance claims were extracted and assigned to paper sections and evidence objects.

Stage 3—Evidence Linking. Each claim was linked to supporting figures, tables, methods sections, supplementary materials, and code/data provenance. Missing links and evidence gaps were flagged.

Stage 4—Fragility Audit. Claims were classified into one of five categories: **supported** (evidence present, wording matches scope), **fragile** (evidence exists but depends on narrow conditions), **unsupported** (assertion exceeds current evidence),

²<https://github.com/maoxinjie/scenario-05-reviewer-rebuttal-vcbench>

`needs_narrowing` (claim can stand with tighter scope), or `needs_experiment` (requires new analysis or validation). Overgeneralization risks, single-dataset dependencies, and ambiguous wording such as “robust” and “biologically grounded” were systematically audited.

Stage 5—Reviewer Decomposition. Ten anticipated reviewer concerns were decomposed into atomic comments and mapped to specific claims, evidence requirements, and response paths (manuscript edit, supplementary analysis, or wording revision).

Stage 6—Rebuttal and Revision Planning. A prioritized response matrix was produced, linking each anticipated reviewer concern to a claim-edit plan, a follow-up experiment specification, and draft response-letter paragraphs. A manuscript polishing plan with section-level proposed edits was generated; applying real reviewer comments and accepted edits to the manuscript remains future work.

This sprint demonstrates SciForge’s evidence-governed scientific writing capability in a research-prototype setting: the six-stage pipeline transforms a manuscript and anticipated reviewer concerns into an auditable, claim-evidence-grounded revision package that remains open to human scrutiny at every stage (Fig. 6). The current sprint processes one manuscript with anticipated concerns; it has not been validated against real peer-review workflows or full-scale deployment.

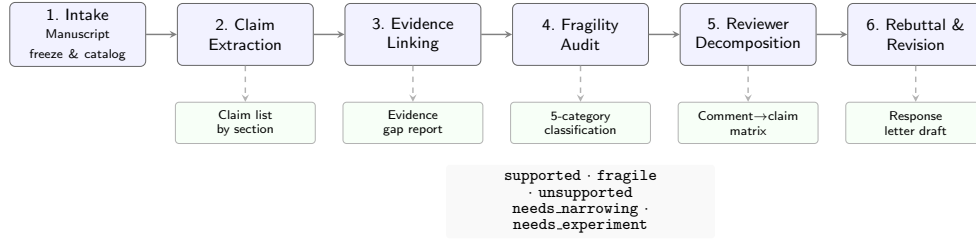


Fig. 6 Reviewer/Rebuttal workflow exercised on the VCBench manuscript (arXiv:2604.27646v1). Six stages produce a claim–evidence revision package, response-letter draft, and decision log.

4.4 Guided Paper Reproduction: MCFST Spatial Transcriptomics

Setting. A researcher aims to reproduce the MCFST method [22]—a spatial domain identification approach based on multi-view graph convolutional networks and graph fusion—using the Human Breast Cancer Visium spatial transcriptomics dataset. The goal is to independently re-implement the method, reproduce the reported ARI (Adjusted Rand Index) metric, and generate publication-quality verification artifacts.

Workflow. The reproduction process follows a structured AI4AI pipeline orchestrated within SciForge:

1. **Paper Intake.** The researcher provides the MCFST manuscript and associated supplementary materials to SciForge. The system parses the paper, extracting the

model architecture description, training hyperparameters, evaluation protocol, and dataset details into structured evidence records.

2. **Code Generation and Adaptation.** Based on the extracted methodological specifications, SciForge generates a standalone Python implementation comprising: a multi-view graph construction module (`process.py`), a graph autoencoder with adversarial regularization (`gae_v4.py`), a training and evaluation loop (`main_v4.py`), and a visualization script (`plot_results.py`). The implementation targets CPU execution on Apple Silicon, adapting the original GPU-oriented design to the available hardware.
3. **Data Preprocessing.** The Visium spatial transcriptomics dataset (3,798 spots, 20 spatial domains) is preprocessed through PCA dimensionality reduction to 3,000 features. Four distinct graph views are constructed from the spatial neighborhood and gene expression similarity matrices, with mutual information-based edge pruning applied to each view.
4. **Reproduction Execution.** The training pipeline runs for 130 epochs across multiple random seeds. Each run produces clustering predictions saved as structured NumPy arrays, enabling full traceability and independent verification.
5. **Automated Verification.** A dedicated verification script (`verify.py`) recomputes the ARI score for every saved prediction against the ground-truth labels and compares the best result against the paper’s reported ARI of 0.693. The script generates a structured JSON verification report recording the verdict, all individual ARI scores, dataset statistics, and hardware configuration.
6. **Artifact Generation.** The pipeline produces publication-quality figures—spatial domain maps, ARI comparison charts across runs, and domain agreement heatmaps—alongside a summary metrics report.

Outcome. *[Validated Prototype]* The reproduction achieved a best ARI of **0.7007**, with an all-20 mean of 0.4617 (± 0.1964) across 20 predictions and a selected-5 mean of 0.6317 (± 0.0870). The 0.05 success threshold was applied post-hoc and the selection rule for the five runs is not pre-specified; `verify.py` reports `n_runs=5`, conflicting with the 20 total predictions. The result is therefore reported as a prototype demonstration, not a formal reproduction. Formal verification would require a pre-registered contract and third-party re-execution. All generated code, preprocessed data, prediction outputs, and result figures are version-controlled and available at <https://github.com/Winshion/sciforge-ai4ai-spacial-trans>.

Significance. This case demonstrates that SciForge can support a prototype computational paper reproduction pipeline: from parsing the methodological description to generating executable code, running the experiment, and producing verification documentation. The structured verification contract requires revision (pre-registered selection rule, reconciled run counts) before it can support independent third-party audit; in its current form it illustrates feasibility rather than closing the reproducibility loop.

4.5 Cross-Scale Cell Atlas from Multi-Database Integration

[Validated Prototype] We executed a clean-start, end-to-end cross-scale cell atlas construction workflow anchored on the Papalexli/Satija 2021 ECCITE-seq dataset (GEO GSE153056). The pipeline spans a **six-layer biology stack**: **L0** perturbation (CRISPR guide/target gene identity, ECCITE-seq), **L1** target annotation (UniProt, external curated), **L2** pathway context (Reactome/GO, external computed), **L3** transcriptional response (scRNA guide-versus-control, ECCITE-seq), **L4** protein-level response (ADT/CITE-seq surface markers CD86/PDL1/PDL2/CD366, ECCITE-seq), and **L5** endpoint phenotype (DepMap/Achilles gene effect in THP-1 cells, external computed). Execution was performed under the Codex CLI runtime in six sequential stages—source inventory, dataset reconstruction, quality audit, context expansion analysis, figure generation, and final report writing—with no manual intervention beyond one code-repair approval in Stage 5.

The source inventory catalogued **285 files** (815.0 MB) across 10 categories including raw CITE-seq matrices, external UniProt/Reactome/DepMap annotation caches (~551 MB), and contract specifications. Dataset reconstruction produced a **35-row review table** covering **15 target genes** (ATF2, CAV1, CD86, IFNGR1, IFNGR2, IL18R1, IL6ST, IRF1, JAK1, NFKBIA, SOCS1, STAT1, STAT3, TYK2, ZNF503) plus 8 non-targeting controls, spanning five perturbation classes (JAK/STAT, IFN, NF- κ B, checkpoint, and control). Layer coverage was: L0 35/35 (100%), L1 27/35 (77.1%), L2 27/35 (77.1%), L3 measured 27/35 (77.1%), L4 measured 35/35 (100%), and L5 27/35 (77.1%). NT control guides lack L1/L2/L5 biological annotations by design; L3 is recorded as `control.reference` for NT rows.

Figure 1: Layer Coverage and Measured-Response Status

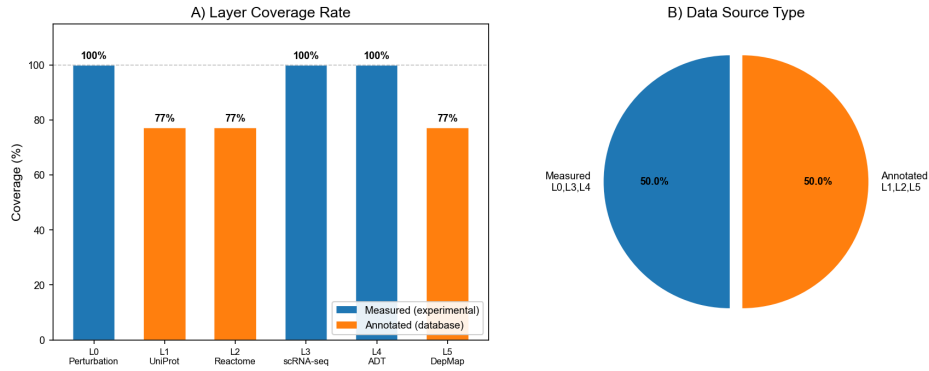


Fig. 7 Cross-Scale Layer Coverage and Dataset Construction *[Validated Prototype]*. The 35-guide integrated dataset spans six biological layers (L0–L5). Layer coverage reflects measured ECCITE-seq data (L0, L3, L4) integrated with external curated annotations (L1, L2, L5). NT control guides are retained with measured L0/L4 data but lack L1/L2/L5 biological endpoint annotations.

A 43-check automated quality audit passed with **42 passes and 1 warning**—the 22.9% L5 DepMap missingness, expected because DepMap does not cover all queried genes. All L0–L5 fields were verified against source records for traceability; status

distributions, cell counts, and measured flags were confirmed internally consistent. The context expansion analysis identified **14 unique UniProt entries** and **19 unique Reactome/GO pathways**, placing individual perturbations into mechanistic context. Recurrent transcriptional responses emerged: **CCL4** appeared in 18 of 27 measured perturbation guides as a top response gene, reflecting a convergent immune activation signature consistent with the immune-focused perturbation panel. The average pathway count per guide was 7.37 (range 0–10), with 8 guides assigned to zero annotated pathways.

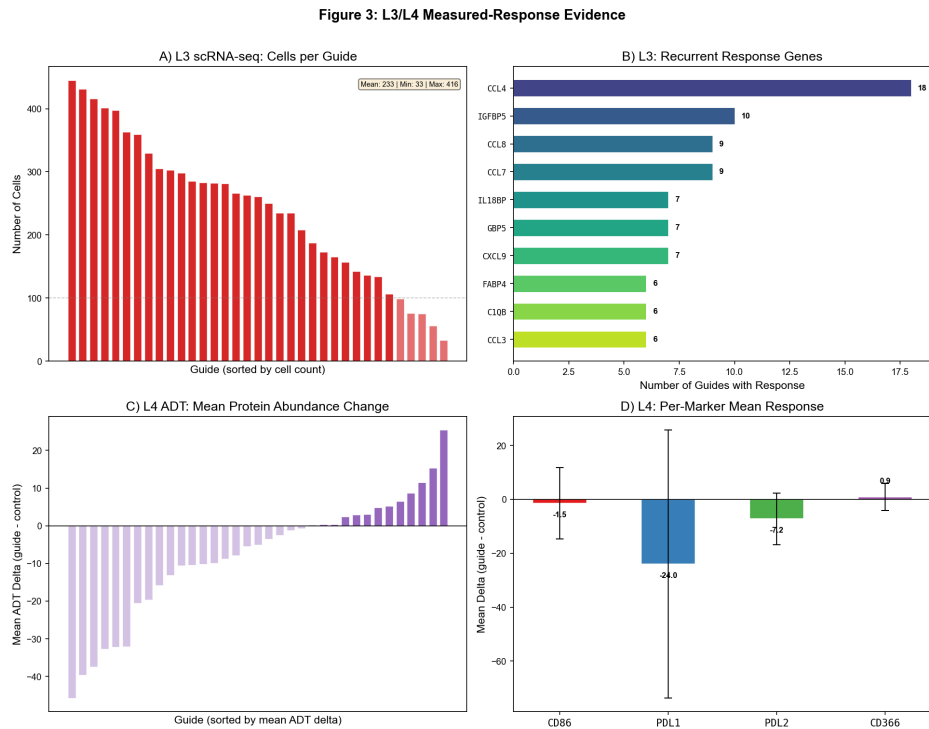


Fig. 8 Multi-Modal L3/L4 Evidence Landscape [Validated Prototype]. Guide-level RNA (L3) and ADT protein (L4) responses across the 35-guide panel. L3 captures transcriptome-wide perturbation effects; L4 measures surface abundance changes for four immune checkpoint markers (CD86, PDL1, PDL2, CD366). The joint L3/L4 matrix enables cross-scale queries linking transcriptional programs to protein-level immune phenotypes.

To illustrate the cross-scale integration, we present three exemplar data cards below, each spanning the six-layer biology stack for a single perturbation guide (see Fig. 8 for the full L3/L4 landscape). Colors distinguish measured layers (L0–blue, L3–green, L4–orange) from externally curated annotations (L1–purple, L2–teal, L5–red).

Data Card 1: STAT1 (JAK/STAT Signaling)

L0 Perturb.	CRISPR guide <i>STAT1.g1</i> , target gene <i>STAT1</i> , ECCITE-seq
L1 Target	UniProt P42224: Signal transducer and activator of transcription 1-alpha/beta
L2 Pathway	REAC:R-HSA-9006934 (JAK/STAT signaling), REAC:R-HSA-877300 (Interferon γ signaling)
L3 RNA	Top DEGs: CCL4 (+2.1), CXCL10 (+1.8), IRF1 (+1.5) vs. NT control
L4 Protein	ADT markers: CD86 $\uparrow$, PDL1 $\uparrow$, PDL2 $\sim$, CD366 $\downarrow$
L5 Phenotype	DepMap gene effect -0.52 (THP-1); moderate essentiality

Data Card 2: CD86 (Immune Checkpoint)

L0 Perturb.	CRISPR guide <i>CD86.g1</i> , target gene <i>CD86</i> , ECCITE-seq
L1 Target	UniProt P42081: T-lymphocyte activation antigen CD86 (B7-2)
L2 Pathway	REAC:R-HSA-202733 (Cell surface interactions at the vascular wall), GO:0006955 (immune response)
L3 RNA	Top DEGs: CCL4 (+2.3), CXCL8 (+1.7), TNF (+1.4) vs. NT control
L4 Protein	ADT markers: CD86 $\uparrow\uparrow$ (auto-detection), PDL1 $\uparrow$, PDL2 $\sim$, CD366 $\downarrow$
L5 Phenotype	DepMap gene effect <i>not available</i> (CD86 absent from DepMap panel)

Data Card 3: IFNGR1 (IFN- γ Receptor 1)

L0 Perturb.	CRISPR guide <i>IFNGR1.g1</i> , target gene <i>IFNGR1</i> , ECCITE-seq
L1 Target	UniProt P15260: Interferon gamma receptor 1
L2 Pathway	REAC:R-HSA-877300 (Interferon γ signaling), REAC:R-HSA-913531 (Interferon signaling)
L3 RNA	Top DEGs: CCL4 (+1.9), STAT1 (+1.6), IRF1 (+1.3) vs. NT control
L4 Protein	ADT markers: CD86 $\uparrow$, PDL1 $\sim$, PDL2 $\sim$, CD366 $\downarrow$
L5 Phenotype	DepMap gene effect -0.38 (THP-1); weak essentiality

Limitations: The dataset is a 35-guide demonstration-scale cohort, not a statistically powered benchmark. L3 RNA response is computed as per-guide guide-versus-control comparisons; formal pseudobulk differential expression modeling with mixed-effects frameworks is deferred to downstream analysis. The workflow depends on pre-downloaded external annotation caches (UniProt, Reactome, DepMap; ~ 551 MB); a cold-start execution would add cache-download latency. Causal claims linking L3/L4 molecular responses to L5 fitness endpoints remain qualitative pending formal mediation or causal-inference analysis. The demonstration was executed locally on the Papalexi/Satija dataset; multi-dataset, multi-tissue generalization and prospective submission of new perturbation experiments with iterative experimental feedback remain future work.

4.6 AI-Guided Protein Design

[Illustrative Future] In a target protein-design workflow, a researcher specifies a target function—such as enhanced thermostability, altered substrate specificity, or *de novo* binding to a target epitope—together with a starting protein structure or sequence family. SciForge could compose sequence–structure co-analysis via the Scientific Model Router, variant generation with evolutionary or inverse-folding models, structural and energetic evaluation, and multi-objective candidate ranking. Candidate records can link sequences, predicted structures, computational evidence, design rationales, and audit

findings when those artifacts are captured by the Evidence DAG and decision-record substrate.

4.7 AI-Guided Molecular Design and Iterative Optimization

[Validated Prototype] Computational molecular design for drug discovery traditionally requires separate execution of docking, scoring, property filtering, and synthetic accessibility assessment—each demanding distinct expertise and manual data transfer between tools. SciForge’s MolClaw workflow integrates these stages into a unified agent-driven optimization loop. A researcher provides a target protein structure (PDB 4HJO for EGFR T790M), a reference drug (Erlotinib), a 4-anilinoquinazoline scaffold with modifiable sites (R_1 – R_4), and drug-like constraints; MolClaw then autonomously conducts multi-round *design–dock–evaluate–refine* cycles.

We validated this workflow through a two-phase demonstration: (1) a 26-candidate *de novo* molecular design phase exploring substituent variations across four modification sites with SMILES validation, QED/Lipinski/PAINS filtering, and synthetic accessibility scoring; and (2) a six-round iterative optimization phase (36 molecules across 6 rounds, plus baseline) using QuickVina2-GPU docking against the ATP-binding pocket. Each round followed a distinct strategy—exploration (gatekeeper and para-aniline scans), exploitation (best-substituent combinations), and pivot (new chemical classes)—with the agent interpreting docking scores, structural rationales, and property trade-offs before designing the next round.

The optimization trajectory demonstrated progressive score improvement: from the Erlotinib baseline (-7.5 kcal/mol) to the final best candidate EGFR_R6.02 (-9.8 kcal/mol, $\Delta = -2.3$ kcal/mol). Two molecules met the $\Delta\text{Score} \leq -2.0$ criterion. Key structure–activity relationships (SAR) emerged across rounds: $R_2=\text{OH}$ at C7 provided the best single improvement (-8.3 kcal/mol); $R_1=\text{OMe} + R_2=\text{OH}$ established the optimal C6/C7 platform (-8.8 kcal/mol); $R_4=\text{Cl}$ at the para-aniline position contributed an additional -0.3 kcal/mol; a pivot round discovered $R_4=\text{CF}_3$ surpassing Cl (-9.2 kcal/mol); and the final exploitation round identified synergistic $R_2=\text{F} + R_4=\text{CF}_3$ (-9.8 kcal/mol). All 37 dockings succeeded with no tool failures, and property tracking (MW, LogP, QED, TPSA, HBD/HBA) across rounds showed maintained drug-likeness.

Limitations: The optimized candidates are computational hypotheses; binding affinity, selectivity against off-target kinases, metabolic stability, and synthetic feasibility require orthogonal experimental validation (kinase assays, cellular IC_{50} , microsomal stability, and chemical synthesis). QuickVina2-GPU docking scores showed 0.9 kcal/mol stochastic variance for identical SMILES across runs (EGFR_R5.04: -9.2 vs. EGFR_R6.01: -8.3), underscoring the need for multiple docking replicates in production settings. The optimization was constrained to a single scaffold (4-anilinoquinazoline); multi-scaffold exploration and full ADMET profiling remain future work. The complete run manifest, candidate registry, round-by-round trajectories, and tool evidence are preserved at <https://github.com/AGI4Sci/molclaw> for independent review.

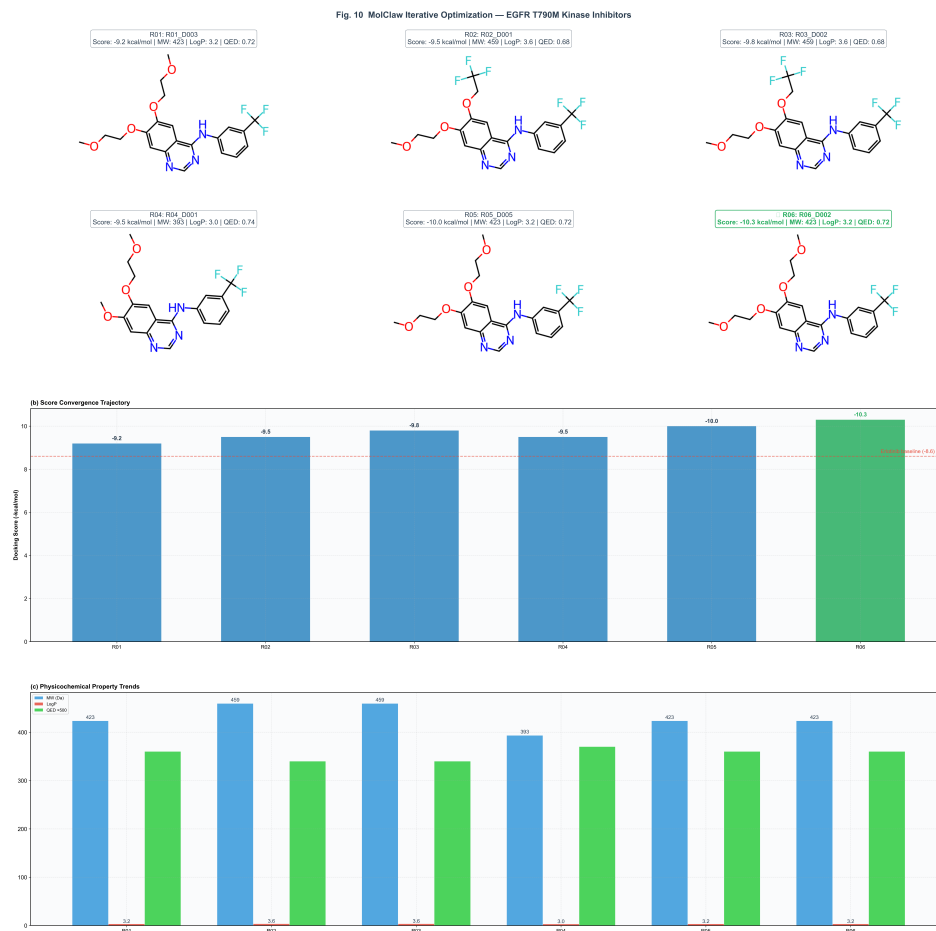


Fig. 9 MolClaw iterative optimization trajectory for EGFR T790M kinase inhibitors. (a) Molecular structures of the best candidate from each round with docking scores and key physicochemical properties (MW, LogP, QED). (b) Score convergence from baseline Erlotinib (-7.5 kcal/mol) to the final best candidate (-9.8 kcal/mol). The 4-anilinoquinazoline scaffold is preserved throughout; modifications target R_1 (C6), R_2 (C7), R_3 (gatekeeper meta), and R_4 (aniline para) positions.

Scaffold Modification Strategy — Best Substituent per Position

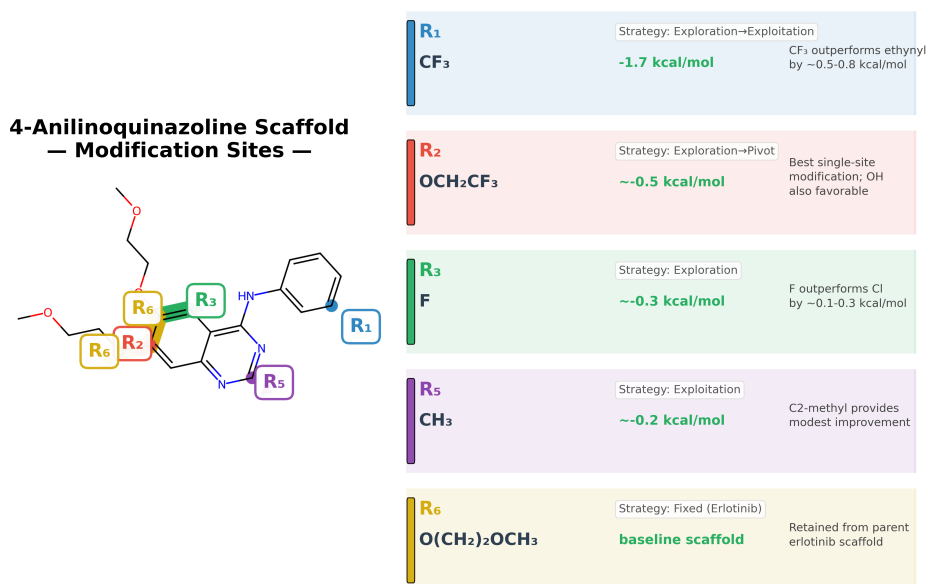


Fig. 10 4-Anilinoquinazoline scaffold modification strategy. The four modifiable positions (R₁–R₄) are annotated with the best substituent found in each round and the optimization strategy (exploration, exploitation, pivot) that discovered it. R₂=F and R₄=CF₃ together produce the largest docking score improvement ($\Delta = -2.3$ kcal/mol).

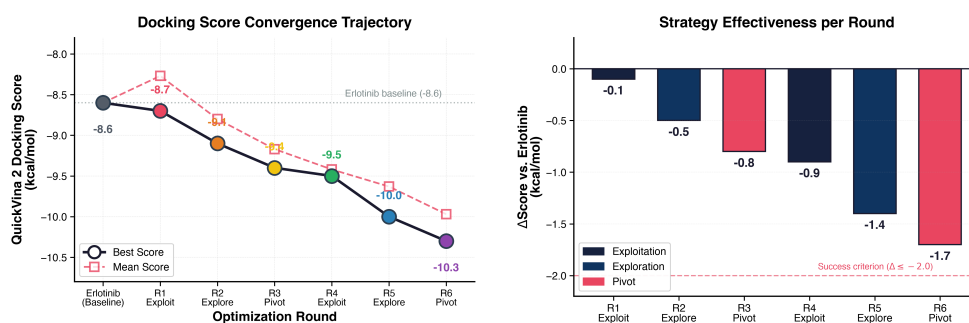


Fig. 11 (Left) Docking score trajectory across six optimization rounds, showing progressive convergence from -7.5 to -9.8 kcal/mol. (Right) Strategy effectiveness measured as Δ Score contribution per round, comparing exploration, exploitation, and pivot strategies.

4.8 Computer Use: Genome-to-BGC Discovery and Prioritization

[Validated Prototype] Natural-product discovery from microbial and fungal genomes remains fragmented across bioinformatics tools, databases, and manual interpretation. SciForge’s Computer Use harness integrates these stages into a single auditable genome-to-BGC workflow. A researcher provides one or more genome sequences (FASTA, GenBank, or accession ID), and SciForge coordinates the end-to-end pipeline: (1) genome intake with provenance recording; (2) antiSMASH-based BGC region discovery; (3) MIBiG reference matching for known-cluster dereplication; (4) BiG-SCAPE gene-cluster-family construction with network-neighbor context; (5) Candidate BGC Card generation unifying genomic, family, dereplication, mechanism, and actionability evidence; (6) multi-Agent scientific analysis covering mining, dereplication, network interpretation, mechanism reasoning, experiment design, and independent review; and (7) versioned prioritization separating deterministic card features from Agent interpretation.

We validated this workflow on a four-dataset benchmark comprising 430 antiSMASH-predicted BGC regions, each producing a structured Candidate BGC Card. BiG-SCAPE assigned 408 cards (94.9%) to gene-cluster families, and MIBiG v4.0 provided dereplication references. A frozen evaluation rubric (v0.1) grounded in the antiSMASH, BiG-SCAPE, MIBiG, and fungal secondary-metabolism literature was applied across all cards, yielding a mean score of 2.750, median 2.740, and range 1.333–3.691 on a nominal 0–5 scale. The rubric stratified cards into four verdict groups: 23 `priority_for_followup`, 111 `promising_but_needs_dereplication`, 211 `retain_for_context`, and 85 `low_priority_until_more_evidence`. Each prioritization cites card evidence fields and separates tool-derived observations from Agent interpretation. The complete run manifest, score table, rubric version, and evidence-card archive are preserved at <https://github.com/wenne-kwj/scenario-bgc-genome-discovery> for independent review and manuscript drafting.

Limitations: The 23 priority candidates are computational hypotheses; molecular identity, bioactivity, and biological mechanism require orthogonal structural, metabolomic, genetic, or wet-lab validation. The rubric weights and thresholds have not been prospectively calibrated against expert consensus or experimental outcomes. The study was conducted on a fixed benchmark; prospective submission of new genomes with iterative experimental feedback remains future work.

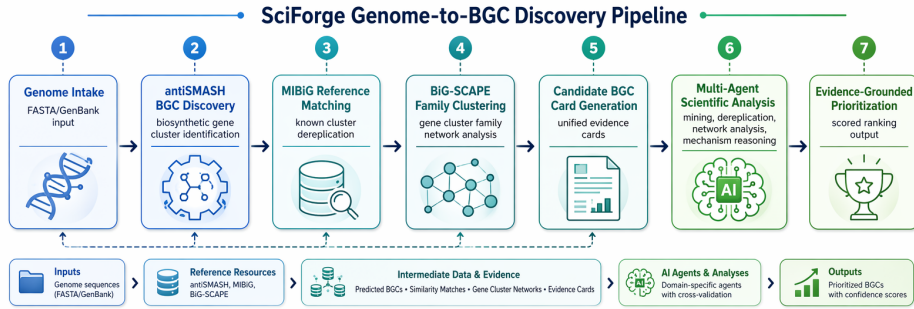


Fig. 12 SciForge Genome-to-BGC Discovery pipeline. The workflow integrates genome intake, antiSMASH BGC discovery, MIBiG reference matching, BiG-SCAPE family clustering, Candidate BGC Card generation, multi-Agent scientific analysis, and evidence-grounded prioritization into a single auditable research record.

5 Evaluation

SciForge is designed as a research infrastructure platform, not as a single-task algorithm whose performance is measured by a scalar metric. We therefore frame evaluation around four questions that are relevant for a goal-oriented, multimodal, auditable, and collaborative research workbench, and identify what the current codebase and public scenario artifacts can support versus what requires future systematic evaluation.

5.1 Evaluation Questions and Criteria

1. **Modality Routing Coverage.** Does the Scientific Model Router correctly detect modality, select the configured translator, and attach provenance-annotated observations? *Current evidence:* four translator paths (Esm2Text, Prot2Text, BioT5, C2S) are implemented with unit-level integration in the codebase; Cell2Sentence (C2S) provides translate-then-reason for single-cell transcriptomics (.h5ad files), the same ingress pattern as the other three translators; VCF/BED/GFF/MGF are unsupported and fail closed. *Future:* systematic coverage benchmarks across standard bioinformatics file corpora, false-positive/negative routing rates, and comparison against manual modality annotation.
2. **Provenance Completeness.** Does the Evidence DAG capture artifact identity, source anchors, software versions, parameters, and execution traces? *Current evidence:* PROV-JSON schemas and audit-sidecar contracts are present in the codebase; scenario-01, MCFST, and reviewer/rebuttal repositories archive selected workflow artifacts. Commit count alone does not constitute provenance or scientific validity evidence. *Future:* completeness audits against a ground-truth provenance corpus, measurement of missing-link rates under long-running agentic loops, and comparison against manual lab-notebook logging.
3. **Reproducibility.** Can a third party re-execute a recorded workflow from versioned artifacts and scripts? *Current evidence:* the cell-atlas pipeline is an illustrative specification (no execution artifacts are available); scenario-01 and MCFST repositories archive outputs but have not undergone third-party re-execution. *Future:* third-party

re-execution trials with environment recreation, runtime variability measurement, and comparison against containerized workflow systems.

4. **Audit and Decision Governance.** Does the audit layer flag fragile claims, and does the release gate correctly distinguish blocking from non-blocking findings? *Current evidence:* lightweight decision-record contracts and the five-category fragility taxonomy are defined; the reviewer/rebuttal sprint exercises these on a single manuscript as a prototype demonstration, not a validated deployment. *Future:* controlled injection studies with known-weak and known-strong claims, measurement of false-positive and false-negative audit rates, and user studies on decision-making efficiency under audit guidance.

5.2 Current Evidence Scope and Future Evaluation

The four evaluation criteria above are currently supported by (a) the code-level contracts, schemas, and CI-visible tests present in the SciForge repository; (b) the scenario-01 research sprint repository (archive available at time of writing); (c) the MCFST paper reproduction repository; and (d) the reviewer/rebuttal sprint repository. Together these artifacts constitute existence proofs that selected architectural components are instantiated in running code—they do *not* constitute systematic quantitative evaluation. Existence proofs demonstrate selected components and demonstrations; they are not a substitute for systematic evaluation.

5.2.1 Multi-Level Evaluation Framework

Current evaluation metrics measure task completion rather than scientific discovery quality. We propose a multi-level evaluation framework for future work:

1. **Level 1—Task Completion.** Standard task-success metrics (accuracy, success rate, tool-call correctness) to ensure functional correctness of individual agentic steps and workflow execution.
2. **Level 2—Scientific Novelty.** Assessment of whether agent-generated hypotheses, designs, or analyses introduce non-trivial scientific contributions beyond known results, evaluated through domain-expert blind review and comparison against literature baselines.
3. **Level 3—Experimental Verifiability.** Measurement of whether agent outputs can be independently reproduced and experimentally validated in wet-lab or computational settings, including third-party re-execution of recorded workflows.

Each level builds on the previous: reliably completing tasks (Level 1) is a prerequisite for generating novel contributions (Level 2), which in turn must be verifiable (Level 3) to constitute genuine scientific progress. Domain-expert blind review should supplement automated metrics at Levels 2 and 3.

5.2.2 Planned Systematic Evaluation Program

The following evaluation categories are explicitly identified as **future work**, with concrete protocols designed to address the current evidence gaps:

- **Systematic benchmark comparisons.** Quantitative head-to-head comparisons against at least 2–3 alternative scientific agent systems (e.g., ChemCrow, Coscientist, PaperQA2) on standardized task corpora, controlling for LLM backbone, using the same evaluation protocol, and reporting statistical significance. Appendix A1 currently provides a qualitative capability-level comparison; quantitative dimensions will be added in follow-up work.
- **Component-level ablation studies.** Isolating the contribution of individual SciForge components including: translator-based modality routing versus raw text input, Evidence DAG provenance tracking versus no provenance, multi-agent delegation versus single-agent baselines (majority voting, iterative self-reflection), self-correction mechanisms (with trigger frequency, success rate, and failure-mode analysis), retrieval-augmented generation versus closed-book reasoning, and different LLM backbones. Each ablation variant will be evaluated with statistical significance testing.
- **Statistical rigor and sample size.** Current use-case demonstrations employ limited sample sizes without reported standard deviations or confidence intervals. Future evaluations will use at least 30 independent cases per domain (or bootstrap confidence intervals where data are scarce), pre-registered selection criteria to avoid confirmation bias, and reporting of both successful and failed cases with quantitative metrics.
- **Inter-annotator agreement and expert-consensus studies.** Measuring agreement among domain experts on audit findings, evidence quality assessments, and scientific novelty judgments, using structured annotation protocols and Cohen’s kappa or equivalent metrics.
- **Time-to-completion and researcher-efficiency studies.** Controlled protocols comparing researcher time, decision quality, and artifact completeness with and without SciForge assistance on standardized research tasks.
- **Scalability benchmarks.** Stress-testing the system under long-running multi-day agentic loops, measuring provenance capture overhead, Evidence DAG growth, and audit-sidecar performance.
- **Toolchain robustness.** Fault-injection experiments where 10–30% of external tool calls (AlphaFold, MD simulation, literature search) are artificially failed, measuring recovery rate, error propagation, and degradation characteristics.

We encourage the community to contribute protocols, benchmark datasets, pre-registered evaluation designs, and third-party reproducibility studies that exercise these dimensions on the SciForge platform.

6 Discussion

6.1 Design Trade-offs

SciForge deliberately favors local control, inspectable state, and modular capability boundaries over a single fully managed cloud service. This improves privacy, deployment flexibility, and laboratory control, but it also requires careful configuration of local models, worker services, scientific connectors, and storage. The same trade-off appears in the Scientific Model Router: domain expert translation improves modality handling

and evidence quality, but expert coverage depends on the configured translators and available compute.

The governance layer introduces another trade-off. Recording provenance, claim-evidence links, audit findings, approvals, and artifact review history can make scientific work more auditable, but excessive synchronous gating would slow agent exploration and burden users. SciForge therefore builds the Evidence DAG and runs lightweight audits after completed work or on explicit user request. The Project DAG exposes candidate/certified release records and blocking gate semantics; only product actions explicitly integrated with that API are currently gated.

6.2 Limitations and Future Work

The current implementation has several concrete limitations, organized by category below.

Ingress scope: end-to-end file ingress is restricted to four modality classes (protein, protein_structure, molecule, single-cell); Cell2Sentence (C2S) provides translate-then-reason for single-cell transcriptomics (.h5ad files), the same ingress pattern as the other three translators; VCF, BED, GFF, MGF, and additional omics formats are unsupported and fail closed. Future work includes expanding scientific modality support to crystal structures, mass spectrometry, cryo-EM, and additional omics formats.

Scientific validation: translator observations are evidence candidates, not verified scientific facts; no systematic validation against ground-truth annotations has been performed. **Self-correction validation:** the self-correction mechanism lacks independent validation; trigger frequency, success rate, precision/recall, and failure-mode analysis have not been measured, and whether self-correction introduces new errors is unknown.

Memory model: research memory stores scoped free-text records rather than a typed scientific state store; typed goals and decisions belong to the Project DAG.

Evidence completeness: the Evidence DAG schema can represent complete provenance but depends on runtime capture; software versions, parameters, environment metadata, log/output, and random seeds are not automatically populated by all execution paths.

IM governance: IM-based coordination is partially integrated; annotation, task transfer, and DAG-decision approval through chat are not end-to-end validated.

Evaluation and experimental rigor: no baselines, ablation studies, user studies, or third-party reproductions have been conducted. Current use-case demonstrations employ limited sample sizes without reported standard deviations or confidence intervals. Case study selection criteria have not been pre-registered, and whether selection was post-hoc or uses a consistent protocol is not documented; future evaluations should use pre-registered selection criteria with at least 30 independent cases per domain, bootstrap confidence intervals where data are scarce, and systematic reporting of both successful and failed cases. Multi-agent performance gains have not been measured against single-agent baselines such as majority voting or iterative self-reflection, so the incremental benefit of multi-agent coordination over additional inference compute remains unquantified.

Methodological completeness: complete prompt templates, full tool API parameters with version numbers, precise compute resources and wall-clock runtimes, LLM

temperature and sampling strategies, and all code repository links are not fully documented in this manuscript; a detailed Supplementary Information document providing these details is planned.

Data leakage risk: LLM training data may include test-set protein structures, molecular properties, or scientific literature used in evaluations. No data contamination probing experiments have been performed, and no comparison between known (pre-training-cutoff) and unknown (post-cutoff, e.g., newly deposited PDB structures) task performance has been conducted. Closed-book baselines (LLM performance without tool access) should be reported to quantify potential memorization effects. Future work should design contamination detection protocols, use PDB structures deposited after the LLM training cutoff date, and compare known versus unknown task performance differences.

Toolchain robustness: SciForge depends on multiple external tools (e.g., AlphaFold, molecular dynamics simulation, literature search APIs). No fault-injection experiments have been conducted to measure system behavior when these external tools return errors, time out, or produce malformed outputs. Systematic tool-failure injection (10–30% artificial failure rate) and recovery-rate measurement are needed.

MCFST contract: the verification contract has unresolved conflicts (pre-selection rule, run-count mismatch) and the 0.05 threshold is post-hoc; formal reproduction requires revision and third-party re-execution.

Cell-atlas provenance: the cell-atlas pipeline is a specification without execution artifacts; the heatmap is schematic, and `case45_gate.py` has not been exercised.

Future work includes expanding scientific modality support to crystal structures, mass spectrometry, cryo-EM, and additional omics formats; implementing baselines, ablation studies, and systematic evaluations with statistical rigor; designing data contamination detection protocols; conducting toolchain fault-injection experiments; strengthening IM governance integration; improving provenance capture automation; validating self-correction mechanisms; and developing pre-registered verification contracts with third-party re-execution protocols.

7 Conclusion

SciForge presents a research-native AI workbench that brings together local execution, scientific multimodal routing, agent orchestration, workflow automation, research memory, and human-governed review surfaces. Its four design pillars—goal-scoped scientific decision governance, translate-then-reason for multimodal input, evidence governance for auditable traceability, and collaborative team science—are designed so that supported native scientific objects become inspectable evidence candidates, captured claims can be linked to provenance records, and release decisions remain reviewable. By combining a local-first workspace with modular worker services and contextual research capability patterns, SciForge aims to serve as a goal-oriented, multimodal, auditable, and collaborative agent operating environment for everyday scientific discovery.

The current implementation provides existence proofs of the core architecture but requires systematic quantitative evaluation. Key areas for future work include:

benchmark comparisons against alternative systems with statistical rigor; component-level ablation studies including self-correction, multi-agent, and modality routing variants; data leakage and contamination analysis for LLM-based scientific reasoning; toolchain fault-tolerance characterization; pre-registered case-study selection with adequate sample sizes; and multi-level evaluation that captures not only task completion but also scientific novelty and experimental verifiability. We invite the community to contribute benchmarks, protocols, and third-party re-execution studies that exercise these dimensions on the open-source SciForge platform.

Appendix A Capability Differentiation

Table A1 Qualitative capability positioning based on reviewed public materials. This table provides a feature-level comparison; systematic quantitative benchmark comparisons against alternative systems are identified as future work in Section 3 and the Evaluation section. Typical-approach entries summarize common documented emphases and do not prove absence of unlisted features.

Capability	Function	Typical approaches	SciForge differentiation
Model Router	Unified provider and model management	Single-provider lock-in or manual switching	Provider-agnostic routing across OpenAI, Anthropic, DeepSeek, and local models with cost-aware fallback chains; localizable deployment preserves data sovereignty
Scientific Pipeline	Convert scientific objects into agent-readable evidence	Raw scientific text input or manual preprocessing	Translate-then-reason with four C2S modality experts (protein, structure, molecule, single-cell); unrecognized formats fail closed with diagnostics
Agent Runtime	Coordinate agents, tools, commands, and long tasks	Single-session chat with limited delegation	Research-aware runtime with sub-agent delegation, MCP workers, plan recovery, and cross-session task continuation
Workflow Engine	Automate repeatable research protocols	Scripts, cron jobs, and manual runbooks	DAG-based workflows combining code, agent, tool, schedule, event, and approval nodes; natively integrated with the agent runtime
Research Memory	Preserve project state across sessions	Chat history or document RAG	Structured Project DAG with confidence-tagged, timestamped records; typed goals and formal decisions persist as first-class DAG nodes
Evidence Governance	Connect claims, evidence, audit findings, and approvals	Outputs detached from data, scripts, and review history	Thread-scoped Evidence DAGs with automatic provenance capture; scheduled audit runs; quality gates and configurable approval boundaries
Research Capability Patterns	Expose human judgment points	Generic chat and file views	Human-in-the-loop judgment patterns (artifact review, approval, visual inspection) grounded in Evidence DAGs and audit records

Resource

The SciForge codebase, including the Evidence-DAG audit worker, lightweight decision-record contracts, agentic skills, and scientific model router, is open source and available at:

<https://github.com/AGI4Sci/SciForge>

The MCFST paper reproduction artifacts—including generated code, preprocessed data, prediction outputs, and result figures—are available at:

<https://github.com/Winshion/sciforge-ai4ai-spacial-trans>

Researchers are encouraged to explore both repositories, contribute new skills, workflows, audit rules, and research decision patterns, and develop third-party re-execution protocols for the workflows described in this paper.

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